

OPEYEMI EBENEZER FATODU

Data Scientist | Machine Learning Engineer | Analytics Engineer

Nigeria | info@yemifatodu.online | +234 703 373 6377 | yemifatodu.online | github.com/yemifatodu | linkedin.com/in/yemi-fatodu

EXECUTIVE PROFILE

Data Scientist and Machine Learning Engineer with a track record of shipping production-grade AI systems — not prototypes. Builds end-to-end, from statistical modeling and ML pipeline design through to deployed, user-facing applications with FastAPI, Streamlit, and modern web stacks. Core strength is bridging technical model output and business decision-making: translating churn probabilities, risk scores, and arbitrage signals into interfaces non-technical stakeholders act on directly. Founder experience architecting a live, multi-service platform from data layer to production deployment.

KEY ACHIEVEMENTS

- 3 production ML/AI systems designed, trained, and deployed end-to-end
- 88.31% accuracy / 0.947 ROC-AUC — best-performing model (GlucoseIQ, XGBoost)
- 86.80% accuracy / 0.864 ROC-AUC — ChurnIQ churn prediction, 10,000-record dataset
- 150 concurrent async requests/cycle across 5 exchanges, 30 symbols — sub-second arbitrage detection
- Founded and built a live multi-service travel platform (flights, hotels, car rentals, eSIM, insurance) serving 3,007 live plans, from data architecture through production deployment

CORE COMPETENCIES

Machine Learning · Data Science · Business Intelligence · Predictive Analytics · ETL & Data Engineering · SQL · Dashboard Development · Model Deployment · API Development · Statistical Analysis · AI Solution Architecture · Full-Stack Development

TECHNICAL SKILLS

Languages: Python (Asyncio), SQL, T-SQL, JavaScript, HTML/CSS

Machine Learning: Random Forest, XGBoost, KMeans, PCA, Regression, Outlier Detection (IQR), Feature Engineering, Class-Imbalance Tuning

Frameworks/Libraries: Pandas, NumPy, Scikit-Learn, Joblib, FastAPI, Pydantic, Uvicorn, Next.js, React

Data & Databases: SQLite, PostgreSQL, Supabase, ETL Architecture, API Integration (CCXT, Telegram Bot API)

BI & Visualization: Streamlit, Tableau, Power BI, Plotly, MS Excel, Jupyter

Cloud & Deployment: Vercel, Stripe, Streamlit Cloud

PROFESSIONAL EXPERIENCE

Founder & Lead Travel Platform Architect / Data Engineer — [Huuboi \(huuboi.com\)](https://huuboi.com)

Remote | March 2024 – Present

- Architected and developed a global travel web application using Next.js, Vercel, and Supabase for flight bookings, hotel reservations, and car rentals across 3,007 live plans
- Integrated third-party APIs and dynamic booking widgets to pull live regional data for flights, hospitality, and rentals
- Designed and implemented telecom billing pipelines featuring global eSIM checkout, automatic data top-ups, and cancellation routing

- Configured payment processing via Stripe and integrated affiliate modules for travel insurance and visa requirement lookups
- Engineered an AI-powered budget planner analyzing user-input parameters to deliver personalized travel financial forecasts

FLAGSHIP PROJECTS

GlucoseIQ | Diabetes Risk Assessment Platform

[Live Demo](#) · [Codebase](#)

- Designed dual ML pipelines (Random Forest + XGBoost) on biostatistical health indicators
- Achieved 88.31% accuracy and 0.947 ROC-AUC (XGBoost)
- Delivered SHAP-based interpretability for clinician-facing risk explanations

ChurnIQ | Predictive Customer Retention Platform

[Live Demo](#) · [Codebase](#)

- Built end-to-end retention intelligence platform on a 10,000-record bank dataset
- RandomForestClassifier (200 estimators, max depth 10) with StandardScaler/OneHotEncoder preprocessing pipeline
- Achieved 86.80% accuracy, 0.864 ROC-AUC; surfaced key churn drivers (Age, Product Holdings, Balance) representing 66% of decision weight
- Documented 44% churn-class recall limitation from dataset imbalance — transparent about model constraints, not just headline accuracy

ArbitrageIQ | Real-Time Multi-Exchange Crypto Arbitrage System

[Live Demo](#) · [Codebase](#)

- Engineered async scanner (asyncio.gather) executing 150 concurrent requests/cycle across 5 exchanges (Binance, KuCoin, Bybit, OKX, Gate), 30 trading symbols
- Built exchange-specific fee calculator subtracting real trading/withdrawal fees, ensuring surfaced spreads represent true executable profit
- Delivered real-time Streamlit + Plotly dashboard with liquidity filtering and Telegram Bot API push alerts

EDUCATION

Applied Data Science Lab (2024) — WorldQuant University, New York, NY, USA

Advanced Diploma in Database Systems / Diploma in Database and T-SQL (2023) — ALISON, Ireland

BSc. Mathematics and Statistics (2006) — University of Calabar, Cross-River State, Nigeria

CERTIFICATIONS

Google Cloud Computing Foundation · Python for Data Science: From Basic to Advanced · Python for Data Science & ML Essential Training · Data Analytics Essentials (Cisco) · Business Intelligence and Analytics (Saylor) · Advanced Excel 2021 · Advanced SQL for Data Science & Time Series · Advanced NoSQL for Data Science